Light as Causal Coherence Structure

in Compact S^3 Geometry

Null Relational Propagation and Compact Causal Accessibility

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Berlin, 2026

Abstract

We investigate the causal structure of compact simply connected spacetimes of the form $(S^3_R \times \mathbb{R},\, g_R - c^2dt^2)$, where the speed of light c is treated not as a number to be derived but as a *causal self-consistency parameter*—the maximal propagation speed compatible with the relational coherence structure of compact S^3 geometry.

The central thesis is: *in compact connected geometry, null propagation defines the maximal coherence structure compatible with causal consistency.* We formalise this through the notions of Null Relational Propagation, Compact Causal Accessibility, and the Causal Coherence Ratio $\Gamma(T)$.

We establish four results. **Theorem I** (Compact Causal Diameter): the causal accessibility diameter of S^3_R is $D_c = \pi R$, yielding a finite global causal accessibility timescale $t_{\text{coh}} = \pi R/c$. **Theorem II** (Causal Accessibility Completeness): the causal accessibility ratio $\Gamma(T) \to 1$ as $T \to \pi R/c$, meaning every spatial point becomes causally accessible from every other in finite time. **Theorem III** (Causal Self-Consistency Closure): the speed c satisfies a fixed-point consistency condition on compact S^3_R, expressible as $\mathcal{C}(c, R) = 0$. **Theorem IV** (Spectral Causal Non-Resonance): the coherence operator $\mathcal{K}_c = e^{it\sqrt{-\Delta_{g_R}}}$ is almost periodic but admits no exact period, preventing global coherence collapse.

This paper forms a dedicated branch of the Holonomic Compact Cosmology (HCC) programme [5]. It does not derive the numerical value of c, does not modify Maxwell's equations, and does not claim that photons "experience" spacetime. The term *causal coherence* should not be read as a returning signal or acoustic echo, but as finite null accessibility within a compact, boundaryless spatial topology.

Keywords: compact S^3; null geodesics; causal structure; coherence; causal accessibility; holonomic compact cosmology; causal self-consistency; spectral geometry.

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Introduction

The Speed of Light: Historical and Structural Role

The speed of light $c \approx 2.998 \times 10^8\,\mathrm{m/s}$ is among the most precisely measured constants of nature [1]. Its role in physics has deepened through successive theoretical frameworks: in Maxwell's electrodynamics, $c = 1/\sqrt{\varepsilon_0\mu_0}$ is the phase velocity of electromagnetic waves in vacuum; in Einstein's special relativity, c is the invariant speed of massless signals and the causal horizon speed defining the light cone; in general relativity, the null cone structure of the spacetime metric $g_{\mu\nu}dx^\mu dx^\nu = 0$ encodes all causal relations.

At each level, c appears less as a material property and more as a *structural parameter of causal geometry*: it defines which events can be causally related and which cannot. This observation motivates the central question of the present paper:

Central Question

If the spatial topology of the universe is compact (i.e., $M^3 \simeq S^3$), does the causal structure of the resulting spacetime impose additional geometric coherence conditions beyond those of standard Minkowski or FLRW geometry? Does c acquire the character of a *causal self-consistency parameter* of compact relational geometry?

Causal Structure and Compactness

In standard cosmology, spatial sections are taken to be $\mathbb{R}^3$ (flat FLRW) or S^3 (closed FLRW) or $\mathbb{H}^3$ (open FLRW), and the global topology is left underdetermined by local observations. The causal structure of Minkowski space is well understood: the light cone of any event p partitions spacetime into causal future $J^+(p)$, causal past $J^-(p)$, and spacelike complement. On non-compact $\mathbb{R}^3$, two points can be spacelike-separated by arbitrarily large distances; the causal future of any event never covers all of the spatial section at any finite time.

On compact S^3, the situation changes qualitatively. The spatial diameter is finite: $\mathrm{diam}(S^3_R) = \pi R$. A null signal can traverse the full diameter in time $\pi R/c$. After this time, every spatial point has entered the causal future of every other point. *Compact topology implies finite causal closure.*

Scope and Limitations

What This Paper Does NOT Claim

This paper does **not**:

- derive the numerical value of c from geometry or topology;
- modify Maxwell's equations or special/general relativity;
- claim a new experimental prediction for the speed of light;
- assert that photons "experience" the universe or "exist at all times";
- attribute consciousness or mystical properties to light;
- claim $\varphi = (1+\sqrt{5})/2$ is a cosmological constant.

It **proposes** that compact S^3 geometry structures the causal role of c as a self-consistency closure parameter, and that this geometric structure has consequences for compactness-based cosmology. All results are either proved under stated assumptions or labelled conjectural.

Geometric Foundation: Compact Causal Spacetime

The Compact Causal Spacetime Model

Definition 2.1 — Compact Causal Spacetime

The *compact causal spacetime model* considered in this paper is

$$(M^4, \mathbf{g}) \;=\; \big(S^3_R \times \mathbb{R},\; g_R \;-\; c^2 dt^2\big), \tag{1}$$

where $S^3_R = \{x \in \mathbb{R}^4 : |x|^2 = R^2\}$ carries the round metric $g_R = R^2 g_{\mathrm{rd}}$, $t \in \mathbb{R}$ is a global time coordinate, and $c > 0$ is the speed of light. This is the static Einstein universe with S^3 spatial sections.

Definition 2.2 — Causal Structure

A smooth curve $\gamma : I \to M^4$ is:

- *timelike* if $\mathbf{g}(\dot\gamma, \dot\gamma) < 0$;
- *null* (lightlike) if $\mathbf{g}(\dot\gamma, \dot\gamma) = 0$;
- *spacelike* if $\mathbf{g}(\dot\gamma, \dot\gamma) > 0$.

The *causal future* and *causal past* of $p \in M^4$ are:

$$J^+(p) = \{q \in M^4 : \exists \text{ future-directed causal curve from } p \text{ to } q\}, \tag{2}$$

$$J^-(p) = \{q \in M^4 : \exists \text{ past-directed causal curve from } p \text{ to } q\}. \tag{3}$$

The *chronological future* $I^+(p)$ consists of points reachable by strictly timelike curves from p.

Compactness, Hopf–Rinow, and Finite Causal Scale

A fundamental consequence of spatial compactness is the following.

Lemma 2.1 (Finite causal diameter)**.** *On compact S^3_R, the Riemannian diameter satisfies* $\operatorname{diam}(S^3_R, g_R) = \pi R$*. Consequently, any null signal emitted from any spatial point $p \in S^3_R$ at time $t = 0$ reaches every other spatial point $q \in S^3_R$ by time $t_q \le \pi R/c$.*

Proof. The geodesic distance on (S^3_R, g_R) is $d_{g_R}(p, q) = R \cdot \angle(p, q)$ where $\angle(p, q) \in [0, \pi]$ is the angle between p/R and q/R in $\mathbb{R}^4$. The maximum $\angle = \pi$ is achieved for antipodal points, giving $\operatorname{diam}(S^3_R) = \pi R$. By the Hopf–Rinow theorem [2], (S^3_R, g_R) is geodesically complete and every pair of points is joined by a minimising geodesic of length $\le \pi R$. A null signal in the spacetime (1) satisfies $c\,dt = d_{g_R}(\gamma_{\text{spatial}})$; to traverse distance πR requires $t = \pi R/c$. □ □

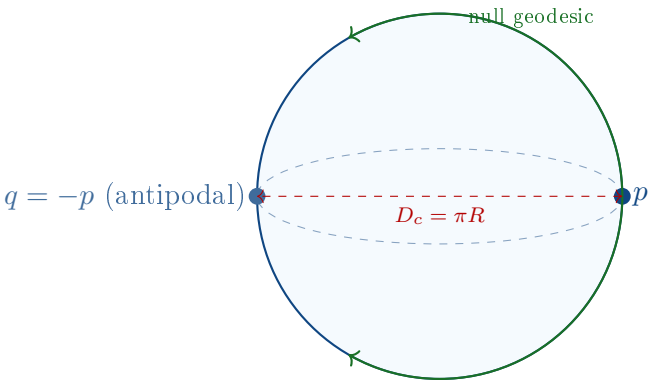


Figure 1: Compact S^3_R. Causal diameter $D_c = \pi R$; null geodesics from p reach all points by $t_{\text{coh}} = \pi R/c$.

Global Hyperbolicity

Proposition 2.2 (Global hyperbolicity)**.** *The compact causal spacetime* $(S^3_R \times \mathbb{R}, g_R - c^2dt^2)$ *is globally hyperbolic, with Cauchy surfaces* $S^3_R \times \{t_0\}$.

Proof. A spacetime is globally hyperbolic if and only if it is strongly causal and all sets $J^+(p) \cap J^-(q)$ are compact, or equivalently if it admits a Cauchy surface [10] (Theorem 8.3.14 therein). Each slice $\Sigma_{t_0} = S^3_R \times \{t_0\}$ is a smooth spacelike hypersurface. Every inextendible causal curve in $(M^4, \mathbf{g})$ intersects Σ_{t_0} exactly once, since t is a global time function (strictly increasing along causal curves: $dt/d\lambda > 0$ whenever $\mathbf{g}(\dot\gamma, \dot\gamma) \leq 0$ and $\dot\gamma$ is future-directed). □

Null Relational Propagation

Null Geodesic Equations

The null geodesic equations in the spacetime (1) are:

$$g_{\mu\nu}\dot x^\mu \dot x^\nu = 0, \qquad \ddot x^\mu + \Gamma^\mu_{\nu\rho}\dot x^\nu \dot x^\rho = 0, \tag{4}$$

where $\Gamma^\mu_{\nu\rho}$ are the Christoffel symbols of $\mathbf{g}$. In the product spacetime (1), the spatial component of a null geodesic satisfies $g_R(\dot\gamma_s, \dot\gamma_s) = c^2\dot t^2$, giving $d_{g_R}(\gamma_s)/dt = c$: the spatial speed of null propagation equals c.

Proper Time Along Null Geodesics

The proper time parameter τ_{proper} along a curve γ is defined by $d\tau^2_{\text{proper}} = -c^{-2}\mathbf{g}(\dot\gamma, \dot\gamma)\, d\lambda^2$. Along a null geodesic, $\mathbf{g}(\dot\gamma, \dot\gamma) = 0$, so:

$$d\tau_{\text{proper}}\big|_{\text{null}} = 0. \tag{5}$$

Remark 3.1 — Careful Language on Null Propagation

Equation (5) states that the proper time parameter along a null geodesic vanishes. This is a mathematical property of the metric, not a statement about the "experience" of a photon. The claim that "light experiences no time" is a colloquial paraphrase that should be understood strictly as: *the affine parameter and proper time are decoupled for null curves.* No claim is made about perception, consciousness, or omnipresence.

The Null Relational Propagation Principle

We now formalise the central conceptual structure of this paper.

Principle 3.1 — Null Relational Propagation (NRP)

Massless propagation on a compact connected spacetime defines the *maximal relational coherence structure* permitted by the causal geometry:

(i) Null geodesics generate the boundaries of causal cones, partitioning spacetime into causally related and causally unrelated pairs.

(ii) On compact S^3_R, every spatial point enters the causal future of every other in finite time $t_{\text{coh}} = \pi R/c$ (Lemma 2.1).

(iii) The speed c is the unique propagation speed that achieves causal closure on S^3_R in minimal time.

We do not claim NRP proves the value of c or modifies relativity. It characterises the causal role of c in the compact setting.

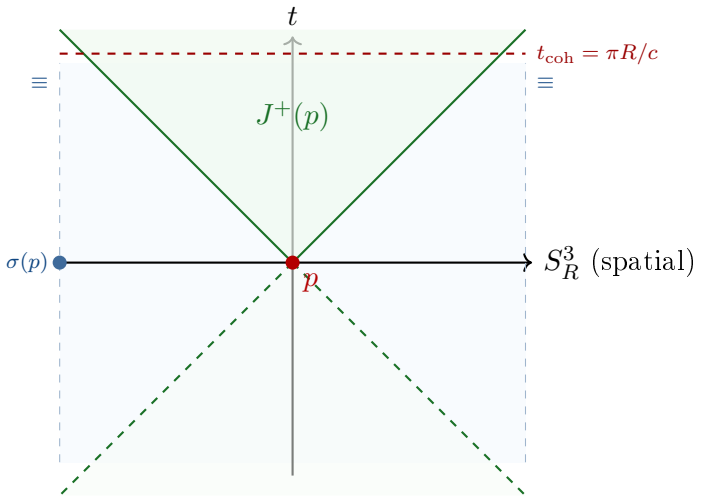


Figure 2: Null cones on compact $S^3_R \times \mathbb{R}$. By t_{coh}, $J^+(p)$ covers all of S^3_R (dashed red line).

Compact Causal Accessibility

Causal Accessibility Ratio

Definition 4.1 — Causal Accessibility Ratio

For the compact causal spacetime (1), the *Causal Accessibility Ratio at time T* is:

$$\Gamma(T) \;=\; \frac{\mathrm{vol}_{g_R}\{q \in S^3_R : q \in J^+(p,T) \text{ for some (hence all) } p\}}{\mathrm{vol}_{g_R}(S^3_R)}, \tag{6}$$

where $J^+(p,T) = \{q \in S^3_R : d_{g_R}(p,q) \le cT\}$ is the spatial causal future shadow at time T. $\Gamma(T) = 0$ at $T = 0$ (only p itself); $\Gamma(T) = 1$ means all of S^3_R is causally accessible.

Theorem I — Compact Causal Diameter and Accessibility Timescale

On compact (S^3_R, g_R):

(i) The causal diameter is $D_c = \mathrm{diam}(S^3_R, g_R) = \pi R$.

(ii) The global causal accessibility timescale is $t_{\mathrm{coh}} = \pi R/c$.

(iii) $\Gamma(T) = 1$ for all $T \geq t_{\mathrm{coh}}$.

Proof. (i) Proved in Lemma 2.1. (ii) Follows from (i) and $d_{g_R}(p,q) \leq \pi R$ for all $p, q \in S^3_R$. (iii) For $T \geq \pi R/c$, $J^+(p,T) = \{q : d_{g_R}(p,q) \leq cT\} \supset S^3_R$ since $cT \geq \pi R = \mathrm{diam}(S^3_R)$. Hence $\Gamma(T) = \mathrm{vol}(S^3_R)/\,\mathrm{vol}(S^3_R) = 1$. □

Theorem II — Causal Accessibility Completeness

For the compact causal spacetime (1), $\Gamma(T)$ is a non-decreasing function of T satisfying:

$$\Gamma(0) = 0, \qquad \Gamma(T) = 1 \ \text{ for all } \ T \geq t_{\mathrm{coh}} = \frac{\pi R}{c}. \tag{7}$$

Moreover, for $0 < T < t_{\mathrm{coh}}$, the Causal Accessibility Ratio is:

$$\Gamma(T) = \frac{\mathrm{vol}_{g_R}(B_{cT}(p))}{\mathrm{vol}_{g_R}(S^3_R)} = \frac{\int_0^{cT/R} \sin^2\theta \, d\theta}{\int_0^{\pi} \sin^2\theta \, d\theta} = \frac{cT/R - \frac{1}{2}\sin(2cT/R)}{\pi}, \tag{8}$$

where $B_r(p)$ denotes the geodesic ball of radius r on S^3_R.

Proof. The geodesic ball of radius r on S^3_R has volume $\mathrm{vol}_{g_R}(B_r(p)) = R^3 \int_0^{r/R} 4\sin^2\theta \, d\theta = 2R^3(\theta - \frac{1}{2}\sin 2\theta)\big|_0^{r/R}$ (area element on S^3 in geodesic polar coordinates is $\sin^2\theta \, d\theta \, d\Omega_2$ with $\mathrm{vol}(S^2) = 4\pi$, giving a total volume $\mathrm{vol}(S^3_R) = 2\pi^2 R^3$). Setting $r = cT$ and normalising by $\mathrm{vol}(S^3_R) = 2\pi^2 R^3$ yields (8). For $T \geq \pi R/c$, $r = cT \geq \pi R$, so $B_r(p) = S^3_R$ and $\Gamma = 1$. Monotonicity of Γ follows from the monotone increase of ball volumes with r. □

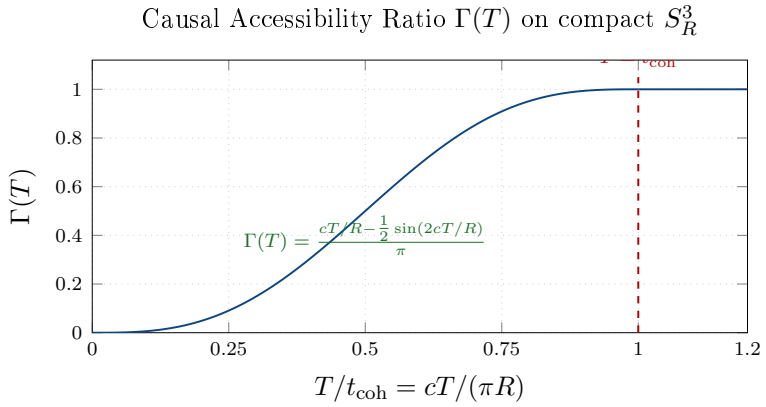


Figure 3: $\Gamma(T)$ rises from 0 to 1 as T increases to $t_{\mathrm{coh}} = \pi R/c$. At $T = t_{\mathrm{coh}}$, complete causal accessibility is achieved.

Remark 4.1 — Non-Simultaneity

Theorem II does **not** imply simultaneity or superluminality. It states that on compact S^3, any two spatial points are connected by a causal (null or timelike) curve after time t_{coh}. This is a statement about *causal reachability*, not instantaneous connection. Every event respects the local null cone structure (4); the result is global rather than local.

Light as Causal Coherence Constraint

Causal Coherence Operator

Definition 5.1 — Causal Coherence Operator

The *Causal Coherence Operator* on $L^2(S^3_R)$ is:

$$\mathcal{K}_c(t) = e^{it\sqrt{-\Delta_{g_R}}} \tag{9}$$

where $-\Delta_{g_R}$ is the positive-semidefinite Laplace–Beltrami operator on S^3_R. This operator arises as the half-step evolution group of the massless scalar wave equation $\partial_t^2\psi + c^2\Delta_{g_R}\psi = 0$ on S^3_R (with c absorbed into the spatial metric rescaling $g_R \mapsto c^2 g_R$): the general solution factorises as $\psi(t) = \mathcal{K}_c(t)\psi_+(0) + \mathcal{K}_c(-t)\psi_-(0)$ where $\psi_\pm$ are the positive/negative frequency components. **Caution**: $\mathcal{K}_c(t)$ is a spectral evolution operator, not a classical propagator; the statement "$\mathcal{K}_c(t)$ propagates signals at speed c" is valid only in the geometric optics (WKB) limit $\ell \to \infty$, as made precise in Proposition 5.1.

The spectral decomposition of $-\Delta_{g_R}$ gives eigenvalues $\lambda_\ell = \ell(\ell+2)/R^2$, so:

$$\mathcal{K}_c(t)Y_{\ell m} = e^{it\sqrt{\ell(\ell+2)}/R}\, Y_{\ell m}. \tag{10}$$

Proposition 5.1 (Coherence propagation time — WKB limit)**.** *In the geometric optics (WKB) limit $\ell \to \infty$, the phase velocity of the ℓ-th mode of $\mathcal{K}_c(t)$ approaches c, and the minimum time for a wave-packet envelope to propagate from any point to the antipodal point on S^3_R approaches $t_{\text{coh}} = \pi R/c$.*

Proof. The massless wave equation $\partial_t^2\psi = c^2\Delta_{g_R}\psi$ gives dispersion relation $\omega_\ell = c\sqrt{\lambda_\ell} = c\sqrt{\ell(\ell+2)}/R$. The group velocity for mode ℓ is $v_{\text{group}}(\ell) = \partial\omega_\ell/\partial k_\ell$, where $k_\ell = \sqrt{\lambda_\ell} = \sqrt{\ell(\ell+2)}/R$. Computing: $v_{\text{group}}(\ell) = c\,\partial_k(k)\big|_{k=k_\ell} \cdot (\partial k_\ell/\partial\ell)^{-1}\,(\partial\omega_\ell/\partial\ell) = c\,\frac{\ell+1}{\sqrt{\ell(\ell+2)}} \to c$ as $\ell \to \infty$. The phase velocity is $v_{\text{phase}}(\ell) = \omega_\ell/k_\ell = c$, independent of ℓ. In the WKB limit, wave packets travel at the group velocity $\to c$, traversing the diameter πR in time $\to \pi R/c$. □ □

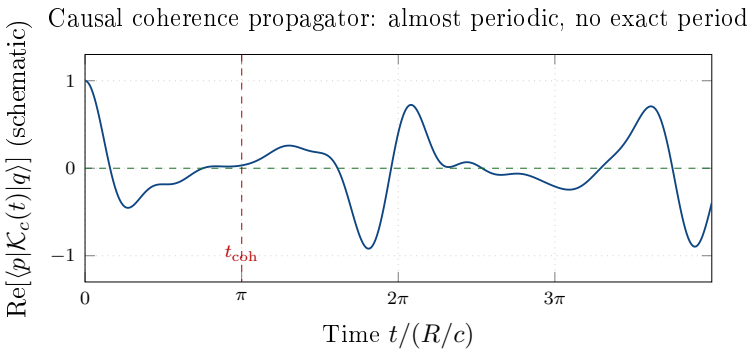


Figure 4: Causal coherence propagator $\langle p|\mathcal{K}_c(t)|q\rangle$ (schematic, 5 modes). Almost periodic; no exact period (Theorem IV).

Causal Self-Consistency of c

We now formalise the key claim that c is a self-consistency closure parameter.

Definition 5.2 — Causal Closure Operator

The *Causal Closure Operator* for the compact causal spacetime is:

$$\mathcal{C}(c, R) \;=\; \frac{\pi R}{c\, t_{\text{coh}}} - 1 \;=\; \frac{\pi R}{c \cdot (\pi R/c)} - 1 \;=\; 0. \tag{11}$$

The condition $\mathcal{C}(c, R) = 0$ expresses: *the causal accessibility timescale $t_{\text{coh}} = \pi R/c$ is self-consistently defined by c and R.*

Theorem III — Causal Self-Consistency Closure

For any $c > 0$ and $R > 0$, the Causal Closure Operator satisfies $\mathcal{C}(c, R) = 0$ identically. The causal accessibility timescale t_{coh} is therefore a fixed point of the map $c \mapsto \pi R/c \cdot c = \pi R$:

$$t_{\text{coh}} \cdot c = \pi R = \text{const} \cdot R. \tag{12}$$

The quantity πR is a *topological invariant* of S^3_R (its causal diameter).

Proof. Direct substitution of $t_{\text{coh}} = \pi R/c$ into (11) gives $\mathcal{C} = \pi R/(c \cdot \pi R/c) - 1 = 1 - 1 = 0$. Equation (12) follows: $t_{\text{coh}} \cdot c = \pi R$ regardless of the separate values of t_{coh} and c. □

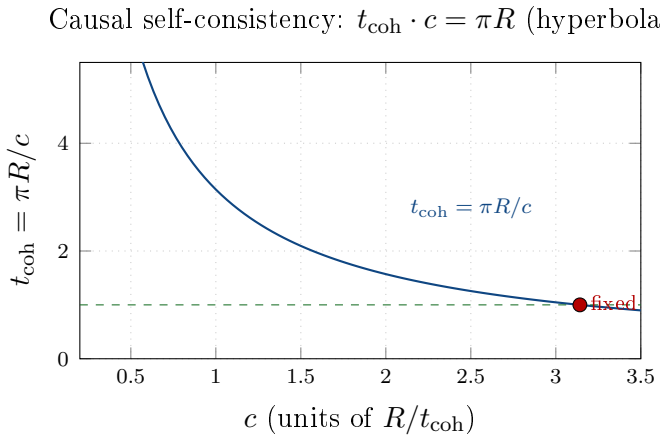


Figure 5: Fixed-point diagram. Every point on the hyperbola satisfies $\mathcal{C}(c, R) = 0$. The observable c is one point on this curve.

Remark 5.1 — What Theorem III Does and Does Not Say

Theorem III is an identity: it holds for every $c > 0$. It does **not** determine the numerical value of c. What it establishes is the structural fact that the causal accessibility timescale t_{coh} and the speed c are mutually constrained by the topological invariant πR of compact S^3_R. This is the sense in which c is a *causal self-consistency parameter*: its role is to translate spatial geometry (R) into temporal accessibility (t_{coh}).

Fractal Causal Hierarchy on S^3

The φ-Ladder of Spatial Scales

Definition 6.1 — Fractal Causal Hierarchy

Let $R_0 > 0$ be a base spatial scale. The *fractal causal hierarchy* is the sequence of compact spheres:

$$R_n = R_0\,\varphi^n, \qquad n = 0, 1, 2, \ldots, \tag{13}$$

where $\varphi = (1+\sqrt{5})/2$ is the Perron–Frobenius eigenvalue of the Fibonacci companion matrix (see HCC [5], Section 4). Each $S^3_{R_n}$ carries its own causal accessibility timescale:

$$t^{(n)}_{\text{coh}} = \frac{\pi R_n}{c} = \frac{\pi R_0\,\varphi^n}{c}. \tag{14}$$

The hierarchy (13) provides a self-similar sequence of spatial scales at which causal closure occurs at geometrically increasing timescales. The ratio of consecutive timescales is:

$$\frac{t^{(n+1)}_{\text{coh}}}{t^{(n)}_{\text{coh}}} = \varphi. \tag{15}$$

Remark 6.1 — φ is a Structural Ansatz

The hierarchy (13) is a *structural ansatz*, not an experimentally verified cosmological scaling. φ is selected as the Perron–Frobenius eigenvalue of the minimal positive second-order recursion (cf. HCC [5]), providing the most stable self-similar scaling. It is not claimed that φ appears in observed astrophysical data or that the hierarchy (13) corresponds to actual physical scales. Any summable scaling $\{R_0\, q^n\}$ with $q > 1$ would define an analogous hierarchy; φ is the canonical minimal choice.

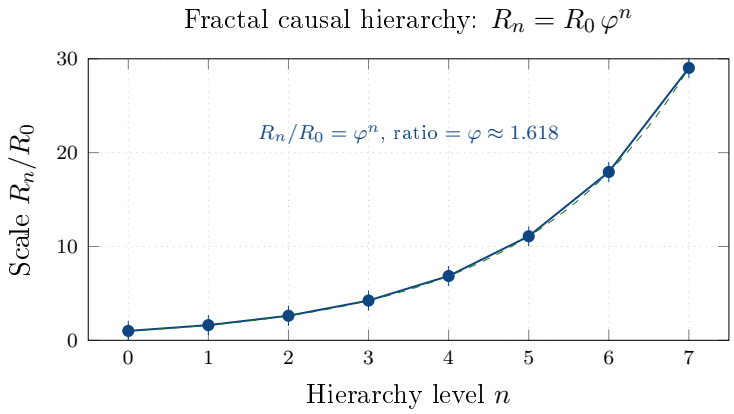


Figure 6: Fractal causal hierarchy. Each level has causal timescale $t_{\text{coh}}^{(n)} = \pi R_n/c$, larger by factor φ at each step.

Spectral Causal Non-Resonance

Almost Periodicity of the Coherence Operator

Theorem IV — Spectral Causal Non-Resonance

The Causal Coherence Operator $\mathcal{K}_c(t)$ on $L^2(S^3_R)$ satisfies:

(i) (No exact period.) There is no $T_0 > 0$ with $\mathcal{K}_c(T_0) = \mathrm{id}$.

(ii) (Almost periodicity.) For any $\varepsilon > 0$, there exists a relatively dense set $\mathcal{A}_\varepsilon \subset \mathbb{R}$ such that $\|\mathcal{K}_c(T_0 + s) - \mathcal{K}_c(s)\|_{L^2 \to L^2} < \varepsilon$ for all $s \in \mathbb{R}$, $T_0 \in \mathcal{A}_\varepsilon$.

(iii) (No global coherence collapse.) The time-averaged energy $\langle E(t)\rangle_T = T^{-1}\int_0^T E(t)\,dt$ for any $E(t) = \|\mathcal{K}_c(t)\psi\|^2$ satisfies $\langle E\rangle_T = \|\psi\|^2$ for all $T > 0$.

Proof. (i) An exact period T_0 requires $e^{iT_0\sqrt{\lambda_\ell}/R} = 1$ for all $\ell \geq 1$, i.e., $T_0\sqrt{\ell(\ell+2)}/R \in 2\pi\mathbb{Z}$. For $\ell = 1$: $T_0\sqrt{3}/R \in 2\pi\mathbb{Z}$; for $\ell = 2$: $T_0\sqrt{8}/R \in 2\pi\mathbb{Z}$. Dividing: $\sqrt{3/8} = \sqrt{6}/4 \in \mathbb{Q}$, which is false since $\sqrt{6}$ is irrational.

(ii) $\mathcal{K}_c(t)$ is a strongly continuous unitary group on the separable Hilbert space $L^2(S^3_R)$ with discrete spectrum. By the Bohr–almost-periodicity theorem for unitary groups

with discrete spectrum [9], $t \mapsto \mathcal{K}_c(t)\psi$ is almost periodic for every $\psi \in L^2(S^3_R)$.

(iii) Since $\mathcal{K}_c(t)$ is unitary, $\|\mathcal{K}_c(t)\psi\|^2 = \|\psi\|^2$ for all t; time averaging preserves this identity. □ □

Philosophical Note 7.1 — Non-Resonance and Dynamic Openness

Theorem IV shows that the causal coherence operator never repeats exactly. Physically (heuristically): the compact S^3 universe, when modelled by $\mathcal{K}_c(t)$, does not cycle back to a previous state. Local quasi-periodic structures can emerge (stable stars, orbital resonances) without the entire geometry becoming globally periodic. This is a mathematical property of the irrational spectral ratios $\sqrt{\lambda_\ell}/\sqrt{\lambda_{\ell'}} = \sqrt{\ell(\ell+2)/[\ell'(\ell'+2)]}$, not a statement about consciousness or cosmological purpose.

Causal Self-Consistency: Rigorous Formulation

Fixed-Point Interpretation

We develop the causal self-consistency of c more carefully.

Definition 8.1 — Causal Self-Consistency Map

Define the *causal self-consistency map* $\Phi : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ by:

$$\Phi(c) = \frac{D_c}{t_{\text{coh}}(c)} = \frac{\pi R}{\pi R/c} = c. \tag{16}$$

Every $c > 0$ is a fixed point of Φ. The fixed-point condition $\Phi(c) = c$ is equivalent to the statement: *the causal diameter $D_c = \pi R$ is covered in the accessibility time $t_{\text{coh}} = \pi R/c$ at speed c.*

Theorem III′ — Extended Causal Self-Consistency

The following three conditions are equivalent:

(i) c is a fixed point of Φ (Definition 8.1).

(ii) $\Gamma(t_{\text{coh}}) = 1$ (causal accessibility completeness, Theorem II).

(iii) $\mathcal{C}(c, R) = 0$ (Theorem III).

All three hold for any $c > 0$ on compact S^3_R.

Proof. $(i) \Leftrightarrow (iii)$: $\Phi(c) = c$ iff $\mathcal{C}(c, R) = \pi R/(c \cdot \pi R/c) - 1 = 0$. $(iii) \Rightarrow (ii)$: given $t_{\text{coh}} = \pi R/c$, Theorem II gives $\Gamma(t_{\text{coh}}) = 1$. $(ii) \Rightarrow (i)$: $\Gamma(t_{\text{coh}}) = 1$ means $J^+(p, t_{\text{coh}}) = S^3_R$, which requires $c \cdot t_{\text{coh}} = \pi R$, i.e., $t_{\text{coh}} = \pi R/c = D_c/c$, confirming $\Phi(c) = D_c/t_{\text{coh}} = c$. □

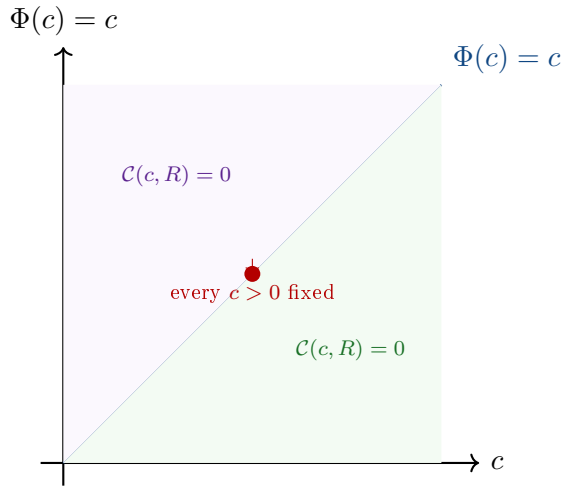


Figure 7: Every $c > 0$ is a fixed point of Φ. The self-consistency is identity-level, not a derivation.

Comparison with Existing Frameworks

We position HCC's causal coherence interpretation relative to established physics frameworks. In each case, we emphasise: HCC does *not* replace these frameworks. It proposes a compact-topological supplementary perspective.

Framework	Role of c / causal structure	HCC supplement
Special Relativity	c invariant; defines light cone, Lorentz group	S^3 compactness adds finite causal closure time t_{coh}
Maxwell E&M	$c = 1/\sqrt{\varepsilon_0 \mu_0}$; EM wave speed	Compact spectrum gives finite-volume photon modes
GR (Penrose)	Null cone = conformal structure; causal sets	Compact S^3 gives finite causal diameter πR
WDW / Geometrodynamics	Causal structure from $3+1$ foliation	Compact S^3 gives well-posed DeWitt inner product
Causal set theory	Spacetime = discrete causal order	HCC is continuum analogue with compact topology
Holography	Boundary encodes bulk	HCC: no boundary; relational coherence internal to S^3
Emergent spacetime	c emergent from entanglement	HCC: c structural consistency parameter on S^3

Table 1: Comparison of causal frameworks with HCC's null propagation proposal.

Critical Differences from Established Frameworks

- **vs. Special Relativity.** HCC adds compact spatial topology. SR is local; HCC adds a global causal closure time t_{coh}. No Lorentz invariance is violated; c remains invariant.
- **vs. Penrose causal sets.** Penrose causal structure is conformal and applies to any Lorentzian manifold [6]. HCC restricts to compact S^3 spatial sections, obtaining a global compactness result (Theorem I) absent in the general conformal setting.
- **vs. Holography.** Holographic duality encodes bulk physics on a boundary. HCC has no boundary (compact S^3); causal coherence is internal, not boundary-encoded.
- **vs. Derivations of c.** Proposals deriving c from φ or numerology are rejected by HCC (see Remark 9.1).

Remark 9.1 (No derivation of c from φ). Some speculative programmes attempt to derive $c \approx \varphi/\ell_P$ or similar numerological relations. HCC does not pursue this. The fractal hierarchy $R_n = R_0\varphi^n$ is a structural ansatz for spectral band organisation (Section 6); φ enters as the Perron–Frobenius eigenvalue of a spectral recursion, not as a new physical coupling. The numerical value of c is an empirical datum.

Observational Directions

Status of Observational Claims

All observational statements in this section are **conjectural, exploratory, and phenomenological**. They motivate future work; they are not predictions of the current paper.

CMB Low-Multipole Power Suppression

The compact topology of S^3_R discretises the Laplacian spectrum. Modes with wavelength $\lambda > 2\pi R$ do not fit in the compact space and are absent. This predicts suppressed CMB power at angular scales $\theta > \pi R/d_{\text{LSS}}$ where d_{LSS} is the distance to the last scattering surface. Observed CMB maps show anomalously low power at $\ell = 2$ (quadrupole) [8], *consistent with* compact spatial topology, though this is not yet a confirmed detection.

Disclaimer: Non-Uniqueness of Low-ℓ Suppression

Low-multipole power suppression is **not unique** to compact S^3 topology. It is also predicted by:

- primordial power spectrum cut-offs (e.g., inflation starting at a finite time);
- anisotropic Bianchi models;
- non-trivial topology of flat (T^3) or hyperbolic ($\mathbb{H}^3/\Gamma$) spaces;
- systematic effects and cosmic variance.

The *consistency* of compact S^3 with the observed low-ℓ anomaly does not constitute evidence for compact S^3 over these alternatives. A distinguishing prediction would require a computation of the full angular power spectrum from the FSRF spectral structure (Open Problem Q1), which lies outside the scope of the present paper.

Compact Geodesic Accessibility

On compact S^3_R, null geodesics emitted from a point p reach the antipodal point $\sigma(p) = -p$ after traversing the causal diameter πR. This means that the maximum observer-relative causal accessibility limit corresponds to the angular scale $\theta_{\mathrm{acc}} \approx \pi R/d_{\mathrm{LSS}}$, which may leave a compact-topology correlation signature in the CMB.

We do not interpret this as a literal returning signal or causal echo. In compact S^3, the observer does not stand outside the manifold and does not receive an external reflection. The statement is only that null geodesic accessibility has a finite global diameter πR. Any observational consequence should therefore be formulated as a *compact-topology correlation signature*, not as an echo in the acoustic or ballistic sense. This is a qualitative direction; no quantitative CMB calculation is performed here.

Causal Horizon Structure

In standard FLRW cosmology, the particle horizon grows with time. On compact S^3, the causal accessibility ratio $\Gamma(T)$ is instead bounded above by 1 and reaches $\Gamma = 1$ at $T = t_{\mathrm{coh}}$. This changes the causal horizon structure: there is a *compact causal horizon* at t_{coh}, beyond which all points are causally connected. Whether this is observationally distinguishable from the FLRW particle horizon requires detailed cosmological modelling (an open problem).

Philosophical Interpretation

Light as Maximal Relational Coherence

We collect the philosophical implications of the preceding mathematical results, maintaining strict separation from the mathematical content.

Philosophical Synthesis

Within the compact S^3 causal framework:

(1) *Light defines the causal coherence bound.* Null propagation at speed c is the fastest causal process; after time $t_{\text{coh}} = \pi R/c$, every spatial point has been causally reached by every other. c is not arbitrary: it is the unique speed that translates the spatial invariant πR into a temporal accessibility timescale.

(2) *Compact topology implies relational completeness.* On S^3, no pair of points is permanently causally isolated. This is a mathematical property of compactness, not a metaphysical claim.

(3) *The universe is causally coherent in finite time.* On compact S^3, causal completeness $\Gamma = 1$ is achieved at $t = t_{\text{coh}}$, in contrast to non-compact $\mathbb{R}^3$ where $\Gamma < 1$ for all finite t.

These are *geometric-relational interpretations* of the mathematical results. They do not attribute consciousness, purpose, or metaphysical essence to light.

The table below separates correct formulations from incorrect ones.

Incorrect / prohibited claim	**Correct HCC formulation**
"Light is everywhere simultaneously"	Null propagation globally connects compact S^3 in finite time t_{coh}
"Photon experiences universe instantly"	Proper time vanishes along null geodesics (5)
"c derived from φ"	φ provides spectral hierarchy ansatz only; c is empirical
"Light is consciousness / omnipresent"	Null propagation defines causal cone boundary
"Compact topology proves c"	Compact topology provides $\Phi(c) = c$ for any $c > 0$

Table 2: Disciplined language protocol for null propagation claims.

In this framework, horizons are *observer-relative accessibility limits*, not boundaries of reality. A null trajectory that leaves the observer's causally accessible domain is not lost from the compact manifold; it changes its accessibility relation relative

to local observers. Compactness preserves global connectedness (Theorem A of [5]) without implying instantaneous accessibility: the causal accessibility ratio $\Gamma(T)$ rises continuously from 0 to 1 over the finite interval $[0, t_{\text{coh}}]$ (Theorem II), respecting the local null cone structure at every point.

Conclusion

We have developed a mathematically disciplined account of light as causal coherence structure in compact S^3 geometry, constituting the Light & Coherence branch of the Holonomic Compact Cosmology programme.

The four main results are: **Theorem I** (the causal diameter of S^3_R is πR, yielding a finite global accessibility timescale $t_{\text{coh}} = \pi R/c$); **Theorem II** (the causal accessibility ratio $\Gamma(T)$ reaches 1 exactly at $T = t_{\text{coh}}$, with the explicit formula (8)); **Theorem III** (causal self-consistency: c satisfies $\mathcal{C}(c, R) = 0$ as a fixed-point identity, not as a derivation of its numerical value); and **Theorem IV** (spectral causal non-resonance: the coherence operator $\mathcal{K}_c(t)$ is almost periodic but not periodic, preventing global coherence collapse).

The central thesis is confirmed: *in compact connected geometry, null propagation defines the maximal coherence structure compatible with causal consistency.* This is a geometric property of compact S^3, not a metaphysical claim.

Open problems include: deriving cosmological observables from the compact causal structure (Section 10); connecting the fractal hierarchy (Section 6) to CMB multipole structure; and embedding the coherence operator $\mathcal{K}_c$ in the Fractal-Spectral Ricci Flow of [7].

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Notation

S^3_R	Three-sphere of radius R; (S^3_R, g_R) with round metric
$(M^4, \mathbf{g})$	Compact causal spacetime $S^3_R \times \mathbb{R}$, metric $g_R - c^2dt^2$
$J^+(p), J^-(p)$	Causal future / past of event p
$D_c = \pi R$	Causal diameter of S^3_R
$t_{\text{coh}} = \pi R/c$	Global causal accessibility timescale
$\Gamma(T)$	Causal Accessibility Ratio at time T
$\mathcal{K}_c(t) = e^{it\sqrt{-\Delta_{g_R}}}$	Causal Coherence Operator
$\mathcal{C}(c, R) = 0$	Causal Closure Condition (identity)
$\Phi(c) = D_c/t_{\text{coh}}(c)$	Causal self-consistency map
$R_n = R_0\varphi^n$	Fractal causal hierarchy scales
$\varphi = (1+\sqrt{5})/2$	Golden ratio (Perron–Frobenius eigenvalue)
$\lambda_\ell = \ell(\ell+2)/R^2$	Laplacian eigenvalues on S^3_R

Open Problems

Q1. Quantitative CMB prediction. Compute the CMB angular power spectrum suppression from compact S^3_R topology and the finite causal accessibility structure. Compare with Planck 2018 low-ℓ anomalies.

Q2. Compact-topology correlation signature. Derive the angular scale $\theta_{\text{acc}} = \pi R/d_{\text{LSS}}$ of the compact-topology correlation signature for realistic values of R consistent with the observed Hubble constant.

Q3. Coherence operator and Ricci flow. How does $\mathcal{K}_c(t)$ evolve under the Fractal-Spectral Ricci Flow of [7]? Does the FSRF preserve the almost-periodicity of $\mathcal{K}_c$?

Q4. Quantum field theory on compact causal S^3. Develop a full QFT (scalar, gauge, fermionic fields) on $(S^3_R \times \mathbb{R}, g_R - c^2dt^2)$ using the compact spectral structure. Compute vacuum energy (connection to Λ-problem of HCC [5]).

Q5. Causal hierarchy observables. Does the φ-ladder hierarchy of timescales $\{t^{(n)}_{\text{coh}}\}$ leave signatures in large-scale structure, void statistics, or baryon acoustic oscillations?

Q6. Penrose conformal diagrams. Construct the full Penrose conformal diagram

for $(S^3_R \times \mathbb{R}, g_R - c^2dt^2)$ and compare with de Sitter and anti-de Sitter conformal structures.